

## Demonstration of Proposed Features

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|----------------------|--------------------------------------------------------------------|
| <b>Date</b>          | 29 June 2026                                                       |
| <b>Team ID</b>       |                                                                    |
| <b>Project Name</b>  | A Comprehensive Measure of Well-Being (HDI Prediction using AI/ML) |
| <b>Maximum Marks</b> | 1 Mark                                                             |

### Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

| S. No | Feature Name                                                 | Description                                                                                                                                          | Status             | Demonstrated | Remarks                                                                              |
|-------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------|--------------------------------------------------------------------------------------|
| 1     | <b>HDI Prediction Engine</b>                                 | Predicts the HDI score using a trained Linear Regression model based on Life Expectancy, Mean Years of Schooling, GNI per capita, and Internet Users | <b>Implemented</b> | <b>Yes</b>   | Verified with multiple test inputs; predictions returned within 1 second             |
| 2     | <b>HDI Category Classification</b>                           | Classifies the predicted score into Low, Medium, High, or Very High HDI category                                                                     | <b>Implemented</b> | <b>Yes</b>   | Tested across all four category ranges with correct labeling                         |
| 3     | <b>Data Pipeline (Preprocessing &amp; Model Persistence)</b> | Cleans and prepares the dataset, then saves the trained model using Pickle for reuse without retraining                                              | <b>Implemented</b> | <b>Yes</b>   | HDI.pkl successfully generated and loaded by the Flask app at startup                |
| 4     | <b>Web-Based User Interface</b>                              | Home page, input form, and result page allowing users to enter indicators and view predictions                                                       | <b>Implemented</b> | <b>Yes</b>   | All pages tested and working with professional theme                                 |
| 5     | <b>HDI Score Comparison &amp; World Map Dashboards</b>       | Bar charts for Top 20/Bottom 10 countries and filterable country cards with Pie/Doughnut charts by HDI category                                      | <b>Implemented</b> | <b>Yes</b>   | Charts render correctly using Chart.js; planning by Spandhana, coding shared by team |
| 6     | <b>HDI Trend Over Years Dashboard</b>                        | Line chart showing HDI trends (1990–2015) by region with a radar chart for multi-indicator comparison                                                | <b>Implemented</b> | <b>Yes</b>   | Verified rendering on Dashboard                                                      |

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|  |  |  |  |  | page; minor load delay noted in Performance Testing report |
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**Feature Implementation Summary:**

|                                 |                                                    |
|---------------------------------|----------------------------------------------------|
| Total Features Proposed         | 6 (covering all 8 implemented functional features) |
| Total Features Implemented      | 6 / 6 (100%)                                       |
| Total Features Demonstrated     | 6 / 6 (100%)                                       |
| Overall Implementation Rate (%) | 100%                                               |